

Reference Summary Table for Methods in ERBOT

Method / Module	Function(s)	Core Idea / Description	Key References
Feature Extraction (TF-IDF + SVD)	er_features_tfidf_svd()	Transforms text to vector space using TF-IDF weighting and truncated SVD (latent semantic features).	Deerwester et al. (1990); Manning et al. (2008)
K-means, Hierarchical, PAM Clustering	er_kmeans_from_X(), er_hclust_from_X(), er_pam_from_X()	Classical vector-space clustering over dense features; evaluates multiple k values via silhouette.	MacQueen (1967); Kaufman & Rousseeuw (1990)
Louvain Community Detection	er_louvain_from_S(), er_louvain_knn()	Graph-based modularity optimization to detect communities from sparse similarity graphs.	Blondel et al. (2008) — 'Fast unfolding of communities in large networks'
Embedding-based kNN Graph	er_embed_knn()	Builds cosine-similarity kNN graph from embedding matrix; clusters via connected components.	Fränti & Sieranoja (2019); Ebraheem et al. (2017)
Edit-distance + Sorted-Neighborhood (SN)	er_mst_or_sn_edit()	Classical deduplication method comparing sorted text pairs within a sliding window, filtered by edit distance threshold.	Hernández & Stolfo (1995); Winkler (1999); Andrzejewski et al. (2023); Ngueilbaye et al. (2021)
MST Graph Pruning	Part of er_mst_or_sn_edit()	Removes long edges beyond median + τ ·MAD distance to produce stable clusters.	Fränti & Sieranoja (2019); Ertöz et al. (2003)
Graph Coloring (GCMER)	er_gc_from_text()	Applies graph coloring with adjustable thresholds to resolve clusters; optimizes via ARI or silhouette.	Papadakis et al. (2016); Neuhoof et al. (2024); GCMER library
Multi-field Similarity Fusion	er_similarity_multifield()	Combines text, categorical, numeric, and year similarities into a weighted sparse similarity matrix.	Christen (2012); Kolb et al. (2012)
Evaluation & Metrics	er_eval_ca_one(), er_silhouette_avg()	Calculates internal (silhouette) and external (ARI, MI, F-measure) clustering metrics for tuning.	Hubert & Arabie (1985); Rousseeuw (1987)